

Software Requirements Specification (SRS)

Attendance Management System

Software Engineering Project Documentation

1. Project Description

The Attendance Management System is a software application designed to simplify the process of recording and managing student attendance. It allows teachers to mark and update attendance, automatically calculates attendance percentages, enables students to view their records, and generates attendance reports while providing secure role-based access.

2. Functional Requirements

Requirement ID	Module/Category	Requirement Description	Priority
FR-01	Authentication & Role Management	The system shall allow authorized users (Administrator, Teacher, and Student) to log in according to their assigned role.	High
FR-02	Attendance Marking	The system shall allow teachers to select a class/subject and mark each student as Present, Absent, or Late for a specific date.	High
FR-03	Attendance Record Management	The system shall save and allow authorized teachers to update attendance records while preventing duplicate attendance for the same student, subject, and date.	High
FR-04	Attendance Percentage Calculation	The system shall automatically calculate and display each student's attendance percentage based on recorded attendance.	High
FR-05	Attendance Viewing	The system shall allow students and teachers to view attendance records, including attendance by subject and date.	High
FR-06	Attendance Reporting	The system shall generate attendance reports based on student, class, subject, or selected date range.	High
FR-07	User Management	The system shall allow administrators to add, update, and deactivate student and teacher accounts.	Medium
FR-08	Class & Subject Management	The system shall allow administrators to create and manage classes and subjects and assign teachers to them.	Medium

Requirement ID	Module/Category	Requirement Description	Priority
FR-09	Student Enrollment	The system shall allow administrators to assign students to their respective classes and subjects.	Medium
FR-10	Attendance Search & Filter	The system shall allow authorized users to search and filter attendance records by student, subject, class, or date.	Medium
FR-11	Attendance Alert	The system shall identify students whose attendance percentage falls below a specified attendance threshold.	Medium
FR-12	Report Export	The system shall allow authorized users to export generated attendance reports in a supported file format.	Medium
FR-13	Password Management	The system shall allow authenticated users to change their account password.	Low
FR-14	Dashboard Summary	The system shall display a dashboard summary containing relevant attendance information based on the user's role.	Low
FR-15	Logout	The system shall allow authenticated users to securely log out of the system and terminate their active session.	Low

3. Non-Functional Requirements

Requirement ID	Category	Requirement Description	Priority
NFR-01	Security	The system shall ensure that users can access only the functions and attendance information permitted by their role, and sensitive data shall be protected during transmission.	High
NFR-02	Performance	The system shall load major pages and attendance information within 3 seconds under normal operating conditions.	High
NFR-03	Reliability	The system shall preserve saved attendance records without data loss and shall perform regular backups of attendance data.	High
NFR-04	Usability	The system shall provide a clear and consistent user interface that allows users to navigate major functions easily.	Medium

Requirement ID	Category	Requirement Description	Priority
NFR-05	Compatibility	The system shall function correctly on commonly used modern web browsers.	Medium
NFR-06	Maintainability	The system shall be designed in modular components so that individual features can be updated and maintained with minimal impact on other components.	Low